

Collective advantage from a minimal record in a quantum information engine

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For a single particle, a quantum information engine can turn measurement fluctuations into transport by raising a barrier behind the particle each time its position is measured. When many particles are present, the barrier still depends on just one number, the position of the leftmost particle, so we let the demon measure that order statistic and nothing else. A demon that resolves every position instead pays a record entropy that grows with particle number, even though the extra distinctions never change where the barrier is placed. We show that the coarser measurement supplies an energy that is bounded independently of particle number, and that this bound yields a ceiling on the record entropy that falls as the filling increases. At the same tilt, the collective engine then delivers 44% more work per recorded nat than independent single-particle engines, each with its own optimized cycle time and each running at equal or greater power per particle. Pauli blocking ends this advantage once the accessible region just above the wall fills. We also evaluate the coherence that the coarse measurement leaves behind, and the cost of placing the barrier imprecisely.

I. INTRODUCTION

Information engines convert measurement and feedback into useful work, and the value of that feedback is limited by the information the demon must eventually erase [1–6]. Feedback enters the second law through the information that the controller acquires [7–13], and the erasure cost itself has by now been measured directly in single-particle experiments [14, 15] and sharpened for reservoirs of finite size [16–18]. When the demon writes to an explicit tape, this accounting becomes a statement about the entropy of the recorded sequence [19–23], which is the form we use throughout.

Quantum information engines have been proposed and realized on several platforms [24–33], among them photonic [34], superconducting [35–37], and nuclear-spin [38] devices. In the quantum case the measurement is more than a source of information, since it can itself supply the energy that the cycle goes on to extract [39–42].

One such engine is purely spatial. Reference [43] places a single particle on a tilted one-dimensional lattice, measures its position, and raises a barrier immediately behind it. Repeating that cycle rectifies quantum measurement fluctuations into transport against the tilt. The contrast with the undriven problem is sharp, because without the demon the particle merely performs Bloch oscillations and travels nowhere over long times [44–47]. Cold atoms realize those oscillations cleanly [48–52], and they persist even without a lattice [53]. Site-resolved experiments have shown, moreover, that few-particle quantum walks in tilted lattices develop correlations and collective Bloch dynamics well beyond the single-particle limit [54]. Extending the transport demon past one particle is therefore

a natural place to ask how much of a many-body state the feedback actually has to resolve.

For many particles, the most direct extension is to measure every position, identify the leftmost particle, and apply the same feedback as before. That extension creates a mismatch at once. A full configuration distinguishes a great many outcomes, yet every configuration sharing a leftmost position calls for the same barrier, so the record grows expensive while the added distinctions change nothing about the action taken. Keeping only what the controller acts on has a long history in information thermodynamics and in partially observable engines [55–58]. In stochastic thermodynamics this is the role played by a sufficient statistic [59], and the information passing between the measured system and its controller obeys a balance of its own [60–62]. Quantum Szilard engines and many-particle demons show, in addition, how particle statistics reshapes information-to-work conversion [24, 63]. What repeated spatial transport adds is a single dynamical model in which the information written to the tape, the measurement backaction, and the collective motion can all be compared.

In the present engine the controlling variable is unusually simple. Wherever the particles happen to sit, the barrier goes immediately below the leftmost one, so the feedback depends on the entire configuration only through that single position. Being the smallest of the occupied site labels, it is an order statistic [64]. We therefore compare two measurements that produce the very same barrier action. The full-configuration measurement resolves the complete configuration, whereas the order-statistic measurement resolves the leftmost position alone and leaves configurations sharing it unresolved. The two consequently differ both in what they record and in the state they leave behind for the next cycle. That difference cannot be reproduced by measuring everything and discarding the rest afterwards, because by then the finer measurement has already changed the quantum state.

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We find that the energy an order-statistic measurement supplies is bounded independently of the number of particles, and that this bound puts a ceiling on the entropy of the record that falls as the lattice fills. At a calibrated ${}^6\text{Li}$ operating point the engine then delivers 44% more work per nat of recorded information than the same number of independent single-particle engines at the same tilt. Each of those runs at the same or greater power per particle. The full-configuration demon, by contrast, stays below that reference at both tilts we examined and for two to eight particles, and about 95% of its ten-particle record resolves distinctions that never change the barrier. The gain is non-monotonic in particle number, since Pauli blocking suppresses the wall once the accessible region just above it fills. A controlled dephasing comparison shows, finally, that the coherence surviving the coarse measurement adds to the work without being necessary for the advantage.

In Sec. II we define the cycle, the two demons, and the single-particle reference. Section III derives the bound on the measurement energy and the entropy ceiling that follows from it. Section IV compares the record cost of the two demons and reports the collective advantage. Section V shows how Pauli blocking limits that advantage. Section VI separates the saving in record cost from the coherence the coarse measurement retains. Section VII examines imprecise barrier placement. Section VIII gives a ${}^6\text{Li}$ lattice realization.

II. SETUP

Consider N spin-polarized fermions on a one-dimensional tilted lattice of M sites, closed on the left by a rigid wall, as shown in Fig. 1(a). Everything below is written in the frame that moves with the wall, so that the lowest site the particles may occupy is always labelled $j = 0$, however far the wall has travelled. We call a state written in this moving frame a *wall-frame state*. The Hamiltonian on the half line is

$$\begin{aligned}\hat{H} &= \hat{K} + \Delta\hat{X}, & \hat{X} &= \sum_{j \geq 0} j\hat{n}_j, \\ \hat{K} &= -J \sum_{j \geq 0} (\hat{c}_{j+1}^\dagger \hat{c}_j + \text{H.c.}).\end{aligned}\quad (1)$$

where J is the nearest-neighbor hopping amplitude and Δ is the energy step between neighboring sites, while $\hat{c}_j$ and $\hat{c}_j^\dagger$ annihilate and create a fermion at site j and $\hat{n}_j \equiv \hat{c}_j^\dagger \hat{c}_j$ counts its occupation. The first term $\hat{K}$ moves particles along the chain. The second is their potential energy, written through the total position $\hat{X} \equiv \sum_j j\hat{n}_j$, so that the product $\Delta\hat{X}$ is an energy and not a change in $\hat{X}$. Throughout the paper we measure energies in units of J , powers in units of $J^2/\hbar$, and time through the dimensionless combination $\tau \equiv Jt/\hbar$. A single dimensionless parameter is then left, the tilt $\alpha \equiv \Delta/J > 0$, and the

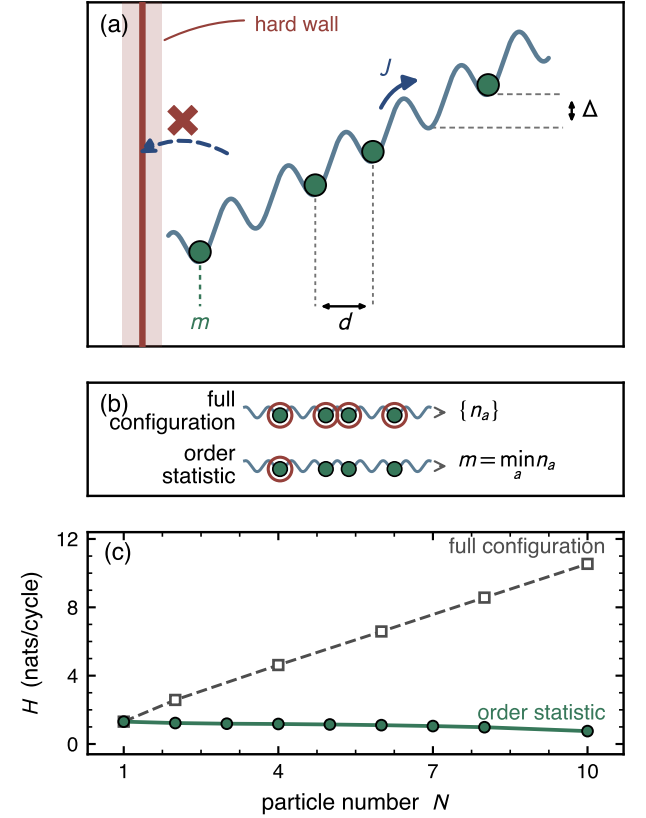


FIG. 1. The full-configuration measurement resolves distinctions that never change the barrier action, while the order-statistic measurement records the variable that controls it. (a) N fermions on a tilted lattice of spacing d and step height Δ , hopping with amplitude J . After each measurement a hard wall is placed below the leftmost particle at m , so that hops across it are forbidden outright rather than merely suppressed. (b) The record each demon takes from the same configuration. The full-configuration demon resolves every occupied site, while the order-statistic demon resolves only the leftmost. (c) Record entropy rate at $\alpha = 0.15$ and $\tau = 1.85$. The full-configuration record grows nearly linearly with particle number. The order-statistic record stays near one nat and decreases as the minimum concentrates. Each curve is evaluated on the histories generated by its own measurement.

numerical results below use Eq. (1) with no interaction between the fermions.

To convert measurement into transport, the engine repeats a four-step cycle.

Step 1, measurement. The demon measures the particle positions, either in full or only through the position of the leftmost particle.

Step 2, feedback. A hard wall is placed on the site immediately below the leftmost particle, so that no particle can hop downward across the new boundary.

Step 3, unitary evolution. The state evolves under Eq. (1) for a time τ .

Step 4, recording. The measurement outcome is appended to the demon's tape, which is kept during the

run and erased once the run is over.

Only Step 2 consumes the outcome, and it consumes very little of it. Writing a configuration as the ordered list $\mathbf{n} = (n_1 < \dots < n_N)$ of occupied sites, the wall is placed below

$$m = n_1 = \min_a n_a, \quad (2)$$

so that two configurations sharing the same m receive the same feedback, whereas two with different m call for the wall in different places. This is the gap the paper exploits, and it separates the two demons we compare. Both follow the same four steps and differ only in Step 1. The *full-configuration (FC) demon* resolves the entire configuration and records $\mathbf{n}$, which leaves the particles in a Fock state. The *order-statistic (OS) demon* resolves only m , and therefore leaves untouched whatever superposition the cycle has built up among configurations that share that minimum. Figure 1(b) contrasts the two records.

Consider now one cycle of the OS demon. A cycle begins immediately after feedback, when the wall-frame state ρ_t has the origin occupied with certainty,

$$\hat{n}_0 \rho_t = \rho_t. \quad (3)$$

Free evolution for a time τ carries this state to $\sigma_t = \mathcal{U}_\tau \rho_t \mathcal{U}_\tau^\dagger$, where $\mathcal{U}_\tau$ is the unitary generated by Eq. (1). The demon then locates the leftmost occupied site with the projectors

$$P_m = \hat{n}_m \prod_{j < m} (1 - \hat{n}_j), \quad p_t(m) = \text{Tr}(P_m \sigma_t), \quad (4)$$

which report the outcome m with probability $p_t(m)$. Each P_m projects onto a whole family of configurations rather than onto a single one, so the outcome alone does not fix the state that the measurement leaves behind. We take the measurement to be of the Lüders type [12, 65], which is the standard choice that disturbs the state no more than the outcome demands. The conditioned state is then $P_m \sigma_t P_m / p_t(m)$, and every coherence internal to the sector survives. The wall is then advanced by m sites and the coordinates are translated by T_m , which returns the first allowed site to the origin. On the half line T_m acts as $T_m |n_1, \dots, n_N\rangle \equiv |n_1 - m, \dots, n_N - m\rangle$ for configurations with $n_1 \geq m$, so it is an isometry on the range of P_m rather than a translation of the whole space, and $T_m^\dagger$ restores the laboratory coordinates. One conditioned cycle therefore reads

$$\rho_{t+1} = \frac{T_m P_m \mathcal{U}_\tau \rho_t \mathcal{U}_\tau^\dagger P_m T_m^\dagger}{p_t(m)}. \quad (5)$$

Because the wall is inserted into a region that is empty by construction, its ideal on-site potential does no mean work while it is relocated, and the translation T_m only resets the frame. In the laboratory frame, every particle keeps the displacement it has accumulated from the moving wall.

The two demons differ in what they must write down. At a fixed conditioned premeasurement state, any exact record Y_t from which the demon can reconstruct the barrier position satisfies

$$H(Y_t | Y_{<t}) \geq H(m_t | Y_{<t}), \quad (6)$$

where m_t is the leftmost position found in cycle t , $H(\cdot | \cdot)$ is the conditional Shannon entropy, and $Y_{<t}$ denotes everything recorded in the cycles before t . Conditioning on that past is essential, because the earlier records fix the state on which the present measurement acts. Recording m_t alone attains that lower bound, and is therefore the minimal record the feedback admits. Any finer record sits above it by whatever the full configuration says beyond its minimum. The two measurements also leave different states behind, so their stationary entropy rates have to be evaluated on their own histories rather than subtracted as though one process produced both.

Because the Hamiltonian is quadratic, both measurements carry a Slater determinant to a Slater determinant, so each demon can be propagated exactly on its own conditioned history. It is enough to follow the N occupied one-particle orbitals, collected as the columns of an $M \times N$ matrix Φ , from which the one-body density matrix $\Gamma = \Phi \Phi^\dagger$ carries every quantity we report below. Appendix A gives that reduction and its many-body checks.

For a run of L cycles, the chain rule gives

$$H(Y_{1:L}) = \sum_{t=1}^L H(Y_t | Y_{<t}). \quad (7)$$

We erase the tape after the run, so the stationary information cost per cycle is the entropy rate $H = \langle -\ln p(Y_t | Y_{<t}) \rangle$ in nats, the mean of the surprisal $-\ln p$ carried by one recorded outcome. We denote the OS and FC rates by $H_{\min}$ and H_{full} .

What the engine produces is most easily seen in the laboratory frame, where the wall has actually moved. Over L cycles it advances by $S_L \equiv \sum_{t=1}^L m_t$ sites in all, carrying the particles up the tilt with it, and the potential energy they gain is

$$W_L = \Delta \left[N S_L + \langle \hat{X} \rangle_{\rho_{L+1}} - \langle \hat{X} \rangle_{\rho_1} \right]. \quad (8)$$

The first term counts the energy stored by moving the wall, since each of the N particles rises by Δ for every site the wall advances. The remaining two terms correct for the cloud rearranging itself relative to the wall, which matters over a finite run but averages away in a stationary state. Dividing by L therefore leaves the stored work per cycle,

$$E = \Delta N \langle m \rangle. \quad (9)$$

That energy comes from the measurement. Unitary evolution conserves the energy of Eq. (1), and the wall does no mean work because it is placed where there is

nothing to push. Averaging over outcomes while keeping no record turns a state σ into $\sum_m P_m \sigma P_m$, and the energy this costs,

$$Q_{\text{meas}} \equiv \text{Tr} \left\{ \hat{H} \left[\sum_m P_m \sigma P_m - \sigma \right] \right\}, \quad (10)$$

is what the demon injects each time it looks. Averaging the feedback translation over outcomes gives

$$Q_{\text{meas}} = \Delta N \langle m \rangle + \langle \hat{H} \rangle_{\rho_{t+1}} - \langle \hat{H} \rangle_{\rho_t}. \quad (11)$$

In a stationary cycle the last two terms cancel, and the balance closes on itself,

$$E = \Delta N \langle m \rangle = Q_{\text{meas}}. \quad (12)$$

Every unit of energy the engine stores is therefore one the measurement supplied, which is what makes the bound of Sec. III so restrictive. Appendix B 1 gives the derivation, and the conditions under which position-diagonal interactions leave it intact.

Running the engine costs what the record will eventually cost to erase, so the ideal efficiency of a cycle is the stored work divided by that work plus the erasure cost of the record [43]. At a fixed erasure temperature, this efficiency rises and falls with a single ratio, the work the engine stores per nat it writes down,

$$R \equiv \frac{E}{H}, \quad (13)$$

at fixed erasure temperature. We report R together with the power per particle $\mathcal{P} = JE/(N\hbar\tau)$, written in script to keep it clear of the outcome probabilities $p_t(m)$. A concrete realization also has control and readout costs. Those enter in Sec. VIII, while R isolates the conversion of measurement energy and recorded information in the ideal cycle.

We compare the collective engine with N independent copies of the single-particle engine at the same tilt. Because a single-particle engine can trade power for work per nat by changing its cycle time, the reference is the envelope

$$R_{\text{ref}}(\mathcal{P}) = \sup_{\tau': \mathcal{P}_1(\tau') \geq \mathcal{P}} R_1(\tau'). \quad (14)$$

A ratio $R_N/R_{\text{ref}} > 1$ means that the collective engine produces more work per recorded nat than every single-particle operating point with equal or greater power per particle. For an arbitrary single-particle cycle time, $R_1(\tau)/R_{\text{ref}}[\mathcal{P}_1(\tau)] \leq 1$, with equality on the optimized envelope. Appendix C constructs the envelope and verifies that the time windows used below contain all power-feasible single-particle points.

III. EXACT RELATIONS FOR THE CYCLE

The OS measurement does more than cut down the number of outcomes the demon has to record. Be-

cause the minimum is a boundary coordinate, a nearest-neighbor hop can change it only when the leftmost ordered particle itself moves. That single observation constrains both the energy the measurement supplies and the entropy of the record it produces. In the same spirit, the Lüders measurement removes only those hopping coherences that connect different values of the minimum, instead of dephasing the cloud throughout its interior. We derive the two constraints in turn.

A. Bound on the measurement energy

We first bound the energy the measurement can inject in a single cycle. That bound does not grow with the number of particles. Every term of Eq. (1) that is diagonal in position commutes with all the P_m , so the energy change in Eq. (10) comes entirely from hopping. What the measurement actually removes is the part of $\hat{K}$ that links different minimum sectors, namely $\hat{K} - \sum_m P_m \hat{K} P_m$, and Q_{meas} is minus its expectation value. In the ordered configuration basis, a nearest-neighbor hop changes the minimum exactly when it moves the first ordered particle. Holding the remaining $N - 1$ coordinates fixed, the positions available to that particle form an open chain, whose hopping block has norm at most $2J$. Summing directly over all the fixed configurations therefore gives

$$|Q_{\text{meas}}| \leq 2J. \quad (15)$$

The bound holds independently of the particle number, and independently of whatever quantum correlations the state carries. It also survives any interaction that is diagonal in position, since such a term commutes with the measurement and therefore survives the subtraction unchanged. On a finite lattice of M sites, which is the geometry we simulate, the exact norm is $2J \cos[\pi/(M - N + 2)]$. Resolving r ordered coordinates rather than one gives $|Q_{\text{meas}}| \leq 2rJ$, and Appendix B 1 proves all of this directly in the ordered occupation basis.

In a stationary engine, Eq. (15) becomes considerably more restrictive than it first appears. Combining it with the balance of Eq. (12),

$$0 \leq E \leq 2J, \quad \langle m \rangle \leq \frac{2}{\alpha N}. \quad (16)$$

However many particles are present, the measurement can supply at most $2J$ per cycle, whereas a wall displacement m has to lift all N of them through Δm each. Adding particles therefore forces the mean advance $\langle m \rangle$ down, simply because the same finite energy has to be shared out more widely.

That ceiling on $\langle m \rangle$ turns into a ceiling on the record. A nonnegative integer variable whose mean is μ can have entropy at most

$$g(\mu) = (1 + \mu) \ln(1 + \mu) - \mu \ln \mu, \quad (17)$$

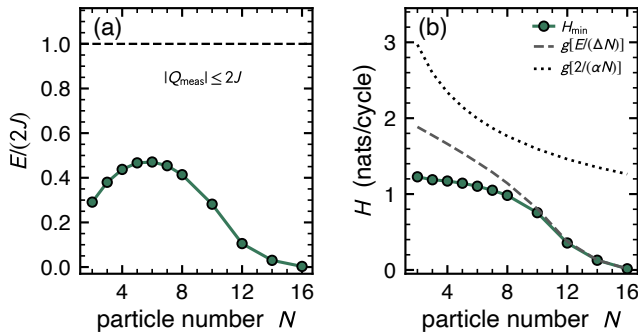


FIG. 2. A boundary measurement supplies only a finite amount of energy even as the particle number grows. (a) Stationary stored work at $\alpha = 0.15$ and $\tau = 1.85$. The dashed line is the exact bound $2J$. (b) The corresponding record entropy rate. The dashed curve is the ceiling obtained from the measured work, $g[E/(\Delta N)]$, and the dotted curve is the general stationary ceiling $g[2/(\alpha N)]$. The numerical points are stationary averages of the OS demon.

a value attained by the geometric distribution of that mean. Setting $\mu = \langle m \rangle$, and using the fact that an entropy rate never exceeds the corresponding single-cycle marginal entropy, Eq. (16) gives

$$H_{\min} \leq g\left(\frac{E}{\Delta N}\right) \leq g\left(\frac{2}{\alpha N}\right). \quad (18)$$

Equation (18) says more than that the record stays bounded as particles are added, since its right-hand side actually decreases, falling asymptotically as $[2/(\alpha N)]\{1 + \ln(\alpha N/2)\}$ at fixed positive tilt. The inequality applies to every stationary state with the finite internal means that Eq. (12) requires. It does not, however, assert that the measured entropy is monotonic in N at every finite particle number.

Figure 2 shows how these two constraints appear in the noninteracting engine. The stored work in Fig. 2(a) never reaches even half of $2J$, rising to 0.47 of the bound at $N = 6$ and falling away on either side. The measurement therefore supplies its largest useful energy at intermediate filling rather than at the largest particle number, and the decline above $N = 6$ is the same suppression that Sec. V later resolves microscopically. These points, and every numerical result that follows, come from stationary trajectories whose lengths, system sizes, and convergence checks are collected in Appendix D.

The two ceilings in Fig. 2(b) behave differently, and the difference identifies what controls the record. The measured entropy approaches the ceiling built from the stationary work as particles are added, reaching 0.65 of it at $N = 2$ and 0.98 at $N = 14$. Beyond $N \simeq 10$ the two are indistinguishable on the scale of the figure, so at large filling the stationary work fixes the record cost to within a few percent. The ceiling built from the $2J$ bound alone falls only from 2.97 to 1.35 nats over the same range,

because it uses no information about the stationary work. The gap between the two curves is therefore the part of the constraint that the dynamics supplies, as opposed to the part that follows from the operator norm alone. The same argument also limits how often the wall moves at all, since $\Pr(m \geq 1) \leq \langle m \rangle$ gives

$$1 - q_N \leq \langle m \rangle \leq \frac{2}{\alpha N}, \quad q_N = \Pr(m = 0). \quad (19)$$

Large filling therefore forces the no-advance probability toward unity even before the detailed stationary state is known.

B. Bound on the minimum record

The bound just derived rests on the energy balance of the repeated engine, and so applies to the stationary state. A second bound holds cycle by cycle, for any state of N fermions whatever, correlated or not. Let a fixed- N fermionic state satisfy the occupied-origin condition $\hat{n}_0 \rho = \rho$, and let U be a quadratic number-conserving single-particle propagator. Then

$$\Pr(m \geq r) \leq \sum_{j \geq r} |U_{j0}|^2. \quad (20)$$

Occupying the origin singles out one particle, and quadratic evolution carries that particle, and only that particle, into the orbital $U|0\rangle$. Pushing every particle past site r therefore costs at least the weight that this one orbital has beyond r , which is the right-hand side of Eq. (20). Appendix B 2 gives the Fock-space proof and extends it to mixed states.

Summing the tails gives $\langle m \rangle \leq \mu_1$, where $\mu_1 = \sum_j j |U_{j0}|^2$ is the displacement of one particle initially at the wall. Energy conservation and $\|K\| \leq 2J$ give $\mu_1 \leq 2/\alpha$. Hence for every conditioned history,

$$H(m | \text{history}) \leq g(\mu_1) \leq g(2/\alpha). \quad (21)$$

The distinction is useful. Occupation of the origin is sufficient to bound the OS record after quadratic evolution. Gaussian closure is needed only to propagate the complete conditioned many-particle dynamics exactly over repeated cycles.

IV. RECORD COST AND COLLECTIVE ADVANTAGE

The FC demon is the natural point of comparison, since at the same tilt and cycle time it applies exactly the same barrier rule and differs only in how much it resolves. Its record, however, grows rapidly with N [Fig. 1(c)], and the reason that extra information buys so little transport is a matter of geometry. How much work a particle contributes depends on how far from the wall it starts,

and that contribution collapses once the starting site lies beyond the distance the dynamics can cover in a single cycle. Appendix B 3 makes this statement precise, and proves it with a Duhamel estimate that runs over the whole propagation history.

Particles far from the wall therefore add many distinctions to a full configuration record while contributing almost nothing to the work that cycle produces. The resulting imbalance is already visible at matched cycle time, where Table I sets the FC entropy and work against the single-particle values at $\alpha = 0.3$. The record cost stays close to additive, whereas the work is strongly sub-additive. At $\tau = 2.5$, for instance, adding four particles multiplies the entropy cost by about 4.3 but the work by only about 1.4, and the two scalings separate further still as the cycle time grows.

The same separation is visible in Fig. 1(c). At $\alpha = 0.15$ and $\tau = 1.85$ the FC entropy rises nearly linearly over the displayed range, while the OS record stays near one nat and decreases as the minimum becomes concentrated near the wall. How much of the FC record is irrelevant to the barrier can be measured on a single history, without setting two different stationary processes against each other. On an FC history we therefore define

$$I_{\text{unused}} = H(\mathbf{n}_t | \mathbf{n}_{<t}) - H(m_t | \mathbf{n}_{<t}). \quad (22)$$

Both terms refer to the same conditioned state and to the same past, so their difference is the information the demon writes down and never uses. Table II shows how that unused fraction grows with particle number, rising from about 0.52 at $N = 2$ to 0.949 at $N = 10$. By ten particles, then, almost the whole configuration record resolves distinctions that never change where the barrier goes.

The discarded distinctions can still change the post-measurement state and later motion. The OS demon must therefore be propagated on its own conditioned histories rather than inferred by subtracting the stationary record costs of two different demons. The comparison in Eq. (22) diagnoses the information mismatch on one history. Whether removing that mismatch can coexist with a collective work response is the question the rest of this section answers.

Our main operating point for the OS demon is

$$\alpha = 0.15, \quad N = 7, \quad \tau = 1.85. \quad (23)$$

At this operating point the engine stores $E = 0.905J$ per cycle while recording $H_{\min} = 1.053$ nats, which amounts to $R = 0.860J$ of work per recorded nat at a power per particle of $\mathcal{P} = 0.0699J^2/h$. Measured against the full single-particle envelope of Eq. (14), this gives

$$\frac{R_N}{R_{\text{ref}}} = 1.440 \pm 0.011, \quad N = 7, \quad (24)$$

where the quoted figure after each value is the half-range across four independent long trajectories, not a standard error. Toward weaker tilt the gain grows further still,

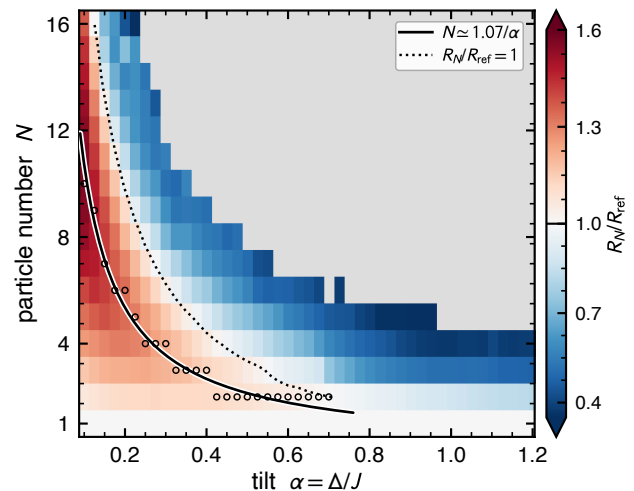


FIG. 3. The collective advantage occupies an intermediate particle-number range. Colour gives the maximum R_N/R_{ref} found over the cycle times considered. The dotted contour marks unit ratio. Black points mark the best particle number found in the scan, and the solid curve $N \simeq 1.07/\alpha$ is an empirical guide through the collective region. Grey cells indicate points where wall motion is too rare to resolve reliably. $N = 1$ is the optimized single-particle control.

reaching 1.569 ± 0.011 at $\alpha = 0.10$ with $N = 10$ and $\tau = 1.85$.

Two quantities already established above combine to produce this ratio. The record cost obeys Eq. (18) and stays near one nat, and the stored work is $E = \Delta N \langle m \rangle$ by Eq. (9), so one wall advance raises all N particles rather than one. Independent single-particle engines have neither. Each pays a separate record for a separate particle, and each decision moves that particle alone. The advantage is the separation between an information cost fixed by a boundary measurement and a displacement shared by the whole cloud.

V. LIMITS SET BY PAULI BLOCKING

The advantage cannot keep growing with particle number. Adding particles at first lets one barrier decision displace a larger cloud while the minimum record stays just as compact, which is the whole source of the gain. Past a certain filling, however, the site just above the wall is increasingly likely to be occupied again once the free evolution ends. The wall then advances less and less often, and both the work and the record activity collapse together.

Figure 3 shows the region of collective gain that results. At each tilt we vary the cycle time and keep the largest ratio found, and the particle number that maximizes the ratio follows an approximate $1/\alpha$ trend across the whole region. A least-squares guide through

TABLE I. The FC demon pays an almost additive record for a strongly sub-additive work and does not reach the single-particle reference. The first two blocks give entropy and work per particle at matched cycle time, relative to one particle, at $\alpha = 0.3$. The third block gives the ratio to the reference after optimizing the cycle time, at $\alpha = 0.3$ and 0.5 .

	$N = 2$	$N = 3$	$N = 4$	$N = 5$	$N = 6$	$N = 7$	$N = 8$
$H_{\text{full}}(N)/[NH_1]$							
$\tau = 2.5$	0.976	0.929	0.896	0.870	0.857	0.844	0.834
$\tau = 4.5$	0.949	0.904	0.877	0.853	0.837	0.820	0.808
$\tau = 6.5$	0.925	0.885	0.860	0.841	0.826	0.811	0.800
E_N/E_1							
$\tau = 2.5$	1.196	1.307	1.380	1.426	1.481	1.512	1.536
$\tau = 4.5$	1.300	1.476	1.608	1.718	1.798	1.860	1.949
$\tau = 6.5$	1.397	1.652	1.842	1.992	2.130	2.233	2.353
R_N/R_{ref} , optimized over cycle time							
$\alpha = 0.3$	0.724	0.584	0.501	0.442	0.399	0.366	0.344
$\alpha = 0.5$	0.705	0.571	0.486	0.429	0.385	0.349	0.326

TABLE II. Unused information in the full-configuration record at $\alpha = 0.15$ and $\tau = 1.85$. $H_m^{(\text{fc})}$ is the entropy rate of the minimum evaluated on the same full-configuration history as H_{full} . The difference is therefore information in the configuration record that is not required to choose the barrier. Values for $N \geq 2$ average four independent 4×10^3 -cycle trajectories. Parentheses give the half-range across those trajectories in the final digit(s). $N = 1$ is exact.

N	$H_m^{(\text{fc})}$	H_{full}	I_{unused}	$I_{\text{unused}}/H_{\text{full}}$
1	1.309	1.309	0.000	0.000
2	1.2432(31)	2.580(20)	1.337(18)	0.5181(31)
4	0.9441(97)	4.623(61)	3.679(51)	0.79576(91)
6	0.749(12)	6.586(80)	5.838(70)	0.88634(67)
8	0.614(15)	8.57(20)	7.96(18)	0.92833(66)
10	0.5351(74)	10.54(17)	10.01(16)	0.94924(62)

the coarse-grid maxima gives $N \simeq 1.07/\alpha$, though neighboring integers are not always resolved from one another. At $\alpha = 0.30$, for example, refined values for $N = 3$ and 4 are 1.2205 ± 0.0050 and 1.2201 ± 0.0067 , which are indistinguishable within their trajectory spreads.

Table III makes two features explicit. First, the relative gain is largest at weak tilt, where the optimized single-particle engine is least economical with the information it records. The absolute work per nat nevertheless stays substantial right across the collective region. Second, the excursion $z(\tau^*)$ varies far less than $1/\alpha$ does, which tells us that the distance explored within one cycle and the number of particles the stationary cloud can hold are set by different things.

The microscopic signature of the suppression is the no-advance probability

$$q_N = \Pr(m = 0) = (UTU^\dagger)_{00}. \quad (25)$$

This identity is exact, because the event $m = 0$ is nothing other than occupation of site zero once the free evolution

TABLE III. Operating points of the OS demon at the particle number the broad scan selects, with the cycle time and the ratio then refined. N_{best} is the particle number with the largest ratio in the scan, τ^* is the cycle time that maximizes the ratio at that N on a grid of 0.025 , and $z(\tau^*)$ is the single-particle Bloch excursion of Eq. (26). Each row uses four independent 1.5×10^4 -cycle trajectories, and the figure after each ratio is their half-range. Repeating every row on $M = 280$ changes no digit shown. Neighboring integer particle numbers can be nearly degenerate, so N_{best} is not resolved from its neighbors at every tilt.

α	N_{best}	τ^*	R_N/R_{ref}	R_N/J	$z(\tau^*)$
0.10	10	1.850	1.569 ± 0.011	0.834	3.69
0.15	7	1.850	1.440 ± 0.011	0.860	3.69
0.20	6	1.750	1.334 ± 0.006	0.871	3.48
0.25	4	1.900	1.281 ± 0.010	0.900	3.76
0.30	4	1.725	1.220 ± 0.007	0.912	3.41
0.40	3	1.750	1.144 ± 0.006	0.946	3.43
0.50	2	1.900	1.099 ± 0.002	0.982	3.66
0.60	2	1.725	1.059 ± 0.005	1.010	3.30
0.70	2	1.650	1.011 ± 0.003	1.020	3.12

has ended. For a Fock state with the particles sitting at the sites n_a , it reduces to the single-particle sum $q_N = \sum_a |U_{0n_a}|^2$. A general conditioned state, however, also carries the off-diagonal elements of Γ , and those coherent contributions are large enough numerically that replacing q_N by the single-particle sum would be wrong.

The natural spatial scale is set by Bloch motion. On the unbounded tilted lattice a particle localized at one site has a Bessel-distributed displacement with argument

$$z(\tau) = \frac{4}{\alpha} \left| \sin \frac{\alpha\tau}{2} \right|. \quad (26)$$

Its maximum $4/\alpha$ is the Bloch-excursion scale [44–47]. The stationary many-particle cloud is not a compact

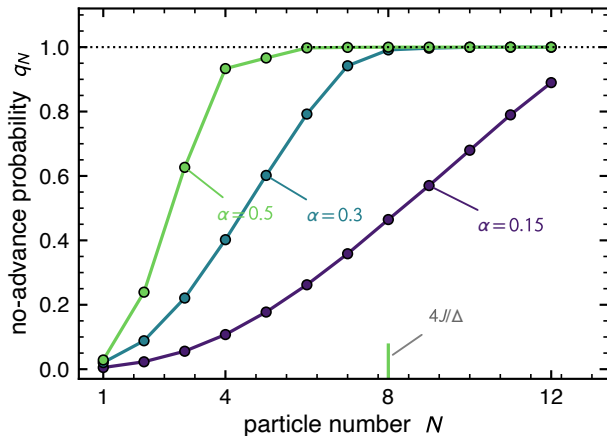


FIG. 4. Pauli blocking suppresses wall motion as the accessible region fills. The no-advance probability q_N is shown against particle number for three representative tilts. The dotted line marks complete blocking, $q_N = 1$. The short mark on the horizontal axis indicates the Bloch-excursion scale $4J/\Delta$ for $\alpha = 0.5$.

wave packet of this width, so $z(\tau)$ is a characteristic excursion scale rather than a strict support bound.

With this interpretation, $z(\tau)$ sets the scale for how far the boundary can reach occupied sites within one cycle. Along the coarse-grid collective maxima for $0.1 \leq \alpha \leq 0.7$, the optimal $z(\tau^*)$ has mean 3.55 and standard deviation 0.25. The cycle therefore uses a similar short-time excursion while the stationary number of particles that the feedback can lift grows roughly as $1/\alpha$.

Figure 4 makes the approach to blocking explicit. As the particle number grows, q_N approaches one and the minimum rarely leaves the origin. The exact inequality in Eq. (19) already forces this trend at large N . Once the advances become very rare, assigning a unique integer stall point depends on how long they are observed. We therefore use the continuous growth of q_N as the microscopic signature of blocking. The useful collective regime lies at intermediate filling, before Pauli exclusion suppresses both wall motion and record production. Strongly tilted lattices are an active experimental setting in their own right, where constrained hopping slows relaxation and fragments the accessible Hilbert space [66–69].

This also clarifies the role of the particle-number optimum. It is a dynamical compromise, not a second entropy theorem. At small N the wall decision is inexpensive but acts on too few particles. At large N the measurement still has a bounded energy input, but the occupied boundary prevents it from translating the cloud. The maximum in Fig. 3 lies between these two limits.

VI. INFORMATION SAVING AND RETAINED COHERENCE

Measurement resolution governs not only what the demon records, but also the quantum state it leaves for the next cycle. An FC position measurement destroys the coherence between configurations that call for the same barrier, whereas the OS measurement leaves that whole sector unresolved and the coherence inside it intact. Both consequences follow from the same choice of measurement, yet they act on the engine in different ways, and the point of this section is to tell them apart.

A. One-body coherence

Coherence enters the work through the off-diagonal part of the one-body density matrix, which the decomposition below isolates exactly. Let Γ be evaluated immediately after feedback, and let U be the single-particle propagator over one free evolution. Write x for the single-particle position operator, diagonal in the site basis with the site index as its eigenvalue. The energy a particle gains over that evolution is then measured by

$$W = \Delta(U^\dagger x U - x), \quad (27)$$

which is simply Δ times the displacement operator. In terms of W , the work of the next unitary step splits exactly into a diagonal and an off-diagonal part,

$$\begin{aligned} E_{\text{unitary}} &= \text{Tr}(W\Gamma), & E_{\text{diag}} &= \text{Tr}[W \text{diag}(\Gamma)], \\ E_{\text{coh}} &= E_{\text{unitary}} - E_{\text{diag}}. \end{aligned} \quad (28)$$

where $\text{diag}(\Gamma)$ keeps only the site populations. The term E_{diag} is thus the work the same populations would give if all coherence were destroyed, and E_{coh} is what the surviving coherence adds on top. All three are evaluated on the same conditioned state, before the next outcome is drawn, so the decomposition carries no measurement shot noise and needs no endpoint correction. Averaged over the stationary cycle, E_{unitary} agrees with the stored work of Eq. (12).

At the seven-particle operating point we find $E_{\text{diag}} = 0.425J$ and $E_{\text{coh}} = 0.479J$, so the coherent term accounts for about 53% of the mean unitary work. For the FC demon, by contrast, the post-feedback state is diagonal in the site basis, and E_{coh} therefore vanishes in every single cycle rather than merely on average.

Figure 5 shows that the coherent share grows with the collective advantage over these operating points, with a correlation coefficient close to 0.9. The contribution is positive throughout most of the weak-tilt regime and can become negative at larger filling. The sign change belongs to the conditioned transport state rather than to a universal notion of coherence as a resource [70, 71]. What coherence can contribute to work is constrained by more than the free energy alone [72–74].

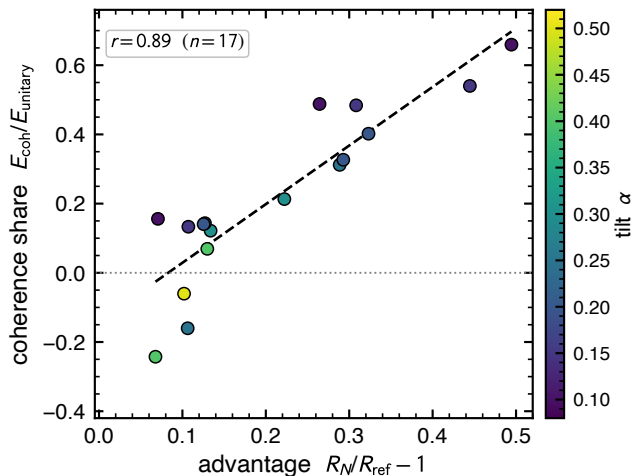


FIG. 5. Coherent work contribution against the collective advantage. Points give the exact coherence share $E_{\text{coh}}/E_{\text{unitary}}$ for the OS demon, coloured by tilt, at operating points with $R_N/R_{\text{ref}} \geq 0.95$. The dashed line is a least-squares fit. For the FC demon the same quantity vanishes identically because the post-measurement one-body density matrix is diagonal in the site basis.

B. Separating coherence from information saving

A stronger test is to remove the coherence while keeping the same coarse record. After measuring m , apply complete position dephasing without recording which configuration occurred,

$$\mathcal{D}(\rho) = \sum_{\mathbf{n}} |\mathbf{n}\rangle \langle \mathbf{n} | \rho | \mathbf{n}\rangle \langle \mathbf{n}|. \quad (29)$$

Such dephasing leaves the recorded variable untouched, while changing both the state and the energy the measurement supplies. For $N = 2$ the resulting wall-frame state is a classical mixture over the separation between the two particles. We evolve that mixture conditioned only on the observed sequence of minima, so that the demon never learns the hidden configuration. Appendix A 3 gives the exact update.

Once the extra dephasing is applied, the OS demon has the same unconditional configuration dynamics as an FC measurement whose detailed record is thrown away. From a given initial ensemble, its mean work is therefore identical to the FC value. What still differs between them is the record, since one demon stores only m while the other stores the whole configuration. Comparing those two cases thus isolates the information saving by itself. Comparing either of them with the Lüders OS demon brings in the retained coherence as well.

The two operating points in Table IV separate the mechanisms. At $\tau = 4.4$, the Lüders OS measurement retains coherence and gives more work than the dephased OS demon, lifting the engine above the single-particle reference. At $\tau = 8$, the fully dephased OS demon itself

TABLE IV. Two-particle comparison at $\alpha = 0.15$. Energy is in units of J and entropy in nats per cycle. Values are obtained from two independent trajectories. The last column uses the single-particle reference at the power of the same row.

τ	Measurement	E	H	R/R_{ref}
4.4	Order statistic	1.371	2.073	1.107
	Dephased order statistic	1.163	2.201	0.885
	Rank one	1.158	3.892	0.498
8.0	Order statistic	1.499	2.432	1.032
	Dephased order statistic	2.061	2.753	1.254
	Rank one	2.054	4.985	0.690

reaches $R/R_{\text{ref}} = 1.254$, showing that the reduced record cost can support a collective advantage even when the post-measurement state is classical in position.

Complete dephasing also changes the measurement backaction by removing additional hopping coherences and can supply energy of its own. At $\tau = 8$ this additional backaction raises the work above the Lüders value. The $2J$ bound of Eq. (15) applies to the Lüders OS measurement alone, whereas complete position dephasing touches the coherences of all N ordered coordinates and therefore falls under the $r = N$ version of the same bound. Together, the three rows separate the reduction in recorded information from the energetic and coherent backaction of the measurement that produces the record.

VII. ERRORS IN BARRIER PLACEMENT

An ideal demon moves the wall by exactly the measured displacement in every cycle. To test what a placement error costs, without at the same time changing what the demon knows, we let the order statistic be measured correctly and record separately whether the wall was misplaced. Specifically, for $m \geq 1$ the wall advances by $m - 1$ rather than by m with probability ϵ , while $m = 0$ leaves it where it is. Since that binary outcome is itself written to the tape, the entropy rate of the record becomes

$$H_{\text{tape}} = H(m_t | \text{past}) + \Pr(m_t \geq 1) h_2(\epsilon), \quad (30)$$

where h_2 is the binary entropy in nats.

Table V gives the stationary performance that follows. The work falls as placement errors accumulate, while the tape entropy rises for two reasons. The misplacement is itself a recorded variable, and earlier errors alter the state handed on to later cycles. The ratio nevertheless stays above the single-particle reference through $\epsilon = 0.10$, and only by $\epsilon = 0.20$ does it drop below.

Table V characterizes a monitored one-site under-placement error, with the misplacement included in the recorded tape.

TABLE V. Effect of a monitored one-site under-placement at the main operating point $\alpha = 0.15$, $N = 7$, and $\tau = 1.85$. H_{tape} includes the minimum record and the recorded misplacement. Values are obtained from four independent 1.5×10^4 -cycle trajectories. Parentheses in the last column give the half-range of the trajectory ratios in the final digit(s).

ϵ	E/J	H_{tape}	R/R_{ref}
0.00	0.905	1.053	1.440(11)
0.02	0.886	1.125	1.319(22)
0.05	0.861	1.193	1.208(8)
0.10	0.817	1.276	1.073(10)
0.20	0.743	1.385	0.898(8)
0.30	0.683	1.444	0.792(4)

VIII. EXPERIMENTAL IMPLEMENTATION

The cycle asks for the same basic cold-atom elements as the single-particle engine, together with two further requirements that are specific to the many-particle case. First, the measurement must locate the leftmost atom without resolving the rest of the configuration. Second, the phase that accumulates while the lattice is frozen has to be controlled. We therefore work through a concrete ^6Li realization in which both requirements are kept explicit.

A. Platform and operating point

The estimates below refer to a concrete set of lattice parameters, fixed here. A vertical ^6Li lattice at wavelength 1064 nm has spacing $d = 532$ nm, and gravity then supplies the step $\Delta/h = 78.64$ Hz. To reach $\alpha \simeq 0.15$, a plane-wave band calculation puts the operating depth at $10.30E_r$, where E_r is the lattice recoil energy with $E_r/h = 29.30$ kHz, and gives $J/h \simeq 524$ Hz. At that depth the gap to the next band is 137 kHz, comfortably larger than every other scale in the problem, and the free evolution of Eq. (23) lasts 0.562 ms. The same band structure gives a next-nearest-neighbor hopping of about 1.1% of the nearest-neighbor value, which is a direct estimate of the leading correction to the ideal Hamiltonian.

B. Preparation

A single-spin-component gas can be prepared as a unit-filled band-insulating segment and imaged with established fermionic quantum-gas microscopy techniques [75–78]. Site-resolved imaging and single-site addressing are by now routine in optical lattices [79–84], and the same microscopes resolve spin and density in fermionic chains [85, 86]. Pauli blocking itself has been observed directly in degenerate fermionic lattice gases [87]. Faster imaging, and imaging below the lattice spacing, shortens the

interrogation each cycle needs [88, 89]. The experiment begins from this prepared Fock state, and after the feed-back cycles are complete, a final site-resolved image gives the laboratory-frame potential energy through Eq. (8).

C. Frozen interrogation

The minimum can be interrogated while hopping is suppressed, which is done by raising the lattice depth. That freezes the spatial occupations, but it does not freeze the tilt, and over a hold of duration t_{fr} the state therefore acquires

$$D_\phi = e^{-i\phi\hat{X}}, \quad \phi = \Delta t_{\text{fr}}/\hbar. \quad (31)$$

For a $360\mu\text{s}$ frozen interval at the operating point, this amounts to 0.178 rad per site. A configuration eigenstate would simply pick up an overall phase and be none the worse for it. The OS measurement, however, retains superpositions of configurations within one sector, so the relative phase between them changes the coherent evolution that follows.

The phase can be removed without changing the measurement outcome. Consider a cycle consisting of free evolution, a rapid intraband ramp to frozen hopping, minimum interrogation, wall feedback, an inverse phase imprint, and restoration of the operating lattice. In the ideal frozen limit the conditional operator is

$$A_m = D_{-\phi} T_m P_m D_\phi \mathcal{U}_\tau = e^{-i\phi N^m} T_m P_m \mathcal{U}_\tau. \quad (32)$$

Since the last factor is only an outcome-dependent global phase, the compensated cycle reproduces Eq. (5) exactly. One could equally compensate the force itself during the interrogation interval. Either way, if the number of threshold questions varies from cycle to cycle, the correction has to be built from the hold duration actually used.

This compensation is essential rather than cosmetic. Keeping the ideal free evolution fixed and adding an uncompensated residual phase of 0.05, 0.10, and 0.178 rad per site per cycle changes the seven-particle ratio to 1.217, 1.017, and 0.746 respectively. The last of these already destroys the advantage. The ideal result is therefore a property of a coherent cycle, and not one that survives an arbitrary frozen hold.

When the frozen interval is implemented by changing the lattice depth, the ramps also enter the energy balance. At exactly frozen hopping, $H_{\text{fr}} = \Delta X$ commutes with the position measurement, so the readout itself has no mean energy backaction. The ramp work is

$$W_{\text{ramp}} = \int dt \dot{J}(t) \left\langle \frac{\partial H}{\partial J} \right\rangle. \quad (33)$$

In the ideal compensated limit this work supplies the kinetic-energy change assigned to the instantaneous measurement in the theoretical cycle. The experimental energy balance must therefore include the lattice ramps together with the probe and the phase-compensation step.

D. Measurement of the order statistic

The demon has to find the leftmost atom without learning anything else, which is the most demanding requirement the protocol places on the apparatus. An adaptive threshold search can locate m using the projectors

$$Q_{\text{empty}}(s) = \prod_{j=0}^s (1 - \hat{n}_j), \quad (34)$$

$$Q_{\text{occupied}}(s) = I - Q_{\text{empty}}(s).$$

Here $Q_{\text{empty}}(s)$ asks whether every site up to s is empty, and a nested sequence of such binary answers multiplies out to the projector P_m . The sequence realizes the OS measurement only if each step is close to the corresponding Lüders projection. Once the binary outcome is fixed, the probe should reveal neither the number of marked atoms nor their positions inside the marked region. Otherwise the environment distinguishes configurations that the ideal measurement leaves coherent.

Dispersive cavity detection offers a route to the required occupied-or-empty response [90–93]. Existing experiments demonstrate high cooperativity, lithium cavity control, and rapid single-atom detection. Cavities already read atoms out nondestructively in the middle of a sequence [94] and count them without resolving which atom is which [95], which establishes the sensitivity and the nondestructive character the threshold search needs. A number-resolved readout is not yet the instrument assumed here, since the count itself separates configurations that share a minimum. The probe must in addition be made insensitive to how many atoms the marked region holds. The feedback must also close within a cycle, and lattice experiments now run real-time control at that speed [96]. The remaining condition is that the binary signal carry negligible configuration information within the accepted sector. This is the distinction between detecting whether a region is occupied and realizing the coarse measurement assumed in Eq. (5).

For a known conditional distribution, the thresholds can be chosen to minimize the expected number of questions. Appendix E gives the exact ordered-search recurrence. At the main operating point the order statistic requires about two threshold questions on average. The number of questions sets the probe exposure, while the record cost is still the entropy rate in Eq. (7).

The same apparatus separates the two demons by how often it must ask. Resolving the full configuration means producing the record whose cost Table II gives as 8.57 nats at $N = 8$. Any protocol built from binary questions then needs at least twelve of them on average, against about two for the order statistic. The question count therefore distinguishes the two measurements operationally, without reference to the model.

E. Backaction and heating

Any probe that scatters light heats the sample, so we estimate what one scattering event costs. Probe recoil is set by the actual 671-nm momentum transfer rather than by the lattice recoil. If the axial momentum transfer is Δk_z in a frozen well of frequency ω_{fr} , one scattering event adds the harmonic-oscillator occupation

$$\delta \bar{n} = \frac{\hbar \langle (\Delta k_z)^2 \rangle}{2m_{\text{at}} \omega_{\text{fr}}}. \quad (35)$$

The excitation probability then follows from this momentum transfer, the frozen-well frequency, and the number of scattering events. Trap-laser noise heats the sample through a separate channel and sets an independent limit on how long the cycle may run [97].

F. Statistics and finite runs

An experiment runs for finitely many cycles, and what it measures over that run differs from the stationary value. A finite run begins from a prepared state and ends with a final position image. Its measured work need not equal the stationary value after only a few cycles. In 512 independent ideal runs of 30 cycles from a packed seven-particle state, terminal imaging gives mean stored energy $1.068J$ per cycle and mean tape surprisal 1.002 nats per cycle. Wall displacement alone would give $0.962J$ per cycle because the internal cloud has not yet lost its endpoint contribution. The finite-run comparison therefore uses Eq. (8) on both the collective and single-particle engines.

The work and the tape surprisal fluctuate together, so the uncertainty of their ratio is best evaluated from the paired totals of each run. Appendix E gives the estimator, including correlations between successive cycles. For the ideal 30-cycle protocol the relative run-to-run spread of the ratio is 0.124. Preparation, detection, and calibration uncertainties enter separately in an experiment.

IX. DISCUSSION AND OUTLOOK

The many-particle engine reveals a separation between the information contained in a quantum state and the information required to control it. Measuring the full configuration resolves an increasing number of distinctions, even though the wall feedback uses only the position of the leftmost particle. The FC record therefore grows rapidly with particle number while much of that information never changes the barrier action. Measuring the order statistic directly removes this growing resolution cost, yet the same wall continues to act on the many-particle cloud. At the calibrated operating point this produces a 44% advantage in work per recorded nat over independent single-particle engines at the same tilt and at the same or greater power per particle.

The exact bounds explain why the minimum record stays so economical. Because the minimum is a boundary coordinate, its Lüders measurement disturbs only the hopping coherences tied to the leftmost ordered particle, and the energy it supplies is therefore bounded independently of N . In a stationary wall frame, that same finite energy has to raise every particle whenever the wall moves, which forces both the mean advance and the entropy ceiling downward as the filling increases. A separate Fock-space argument shows that the transient bound follows from the occupied-origin condition for arbitrary correlated fermionic states. Gaussian closure enables exact propagation of the repeated noninteracting dynamics.

Fermi statistics then decides how far the collective response can be pushed. At low filling a single barrier decision acts on several particles at once. As the accessible region just above the wall fills up, however, Pauli blocking makes the first allowed site ever more likely to be occupied again, and the wall moves less and less often. The result is a maximum at intermediate particle number, beyond which both transport and record production are suppressed. That optimum follows an approximate $1/\alpha$ trend over the collective region. The onset of very rare wall motion is described more directly by the continuous no-advance probability than by a universal integer stall point.

Measurement resolution also governs the state that enters the next cycle. The FC measurement removes the coherence between configurations calling for the same feedback, whereas the OS measurement leaves that sector unresolved. The one-body coherence it retains can contribute substantially to the work at weak tilt. The dephasing comparison shows that the reduced record cost can already support a collective advantage in a position-diagonal state, while retained coherence provides an additional work contribution. These two consequences of the coarse measurement can be separated by changing the measurement backaction while holding the recorded variable fixed.

The experimental implementation makes both distinctions concrete. Freezing the hopping does not freeze the phase that the tilt generates, so a repeated coherent cycle calls for phase compensation, or for some equivalent modification of the propagator. Locating the leftmost atom without destroying the useful coherence likewise calls for a threshold measurement, one that reveals whether a region is occupied without resolving the configuration inside it. These conditions provide a direct experimental test of how measurement resolution affects the engine.

Interactions provide a natural extension. Translation-invariant density interactions between spinless fermions preserve the measurement-energy bound while changing the stationary dynamics, as summarized in Appendix F. Multicomponent fermions and soft-core bosons add further interaction channels [54, 66], and existing cold-atom transport experiments provide settings in which the same feedback principle can be tested [98–100]. The broader question is how information cost and measurement back-

action constrain collective transport once correlations arise from interactions as well as quantum statistics.

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DATA AVAILABILITY

The data and the code that generates them are available upon reasonable request from the first author.

Appendix A: Gaussian reduction and its validation

Because the Hamiltonian is quadratic, every conditioned Slater state can be followed through its occupied one-particle orbitals. We collect the N orthonormal orbitals in an $M \times N$ matrix Φ , with one-body density matrix $\Gamma = \Phi\Phi^\dagger$. If $A_m = \{0, \dots, m-1\}$, the probability that the first m sites are empty is

$$S_m = \det(I - \Gamma_{A_m}), \quad p(m) = S_m - S_{m+1}. \quad (\text{A1})$$

These are the standard determinantal occupation identities for free fermions [101–103]. The same correlation matrix also carries the reduced density matrices of a free-fermion state [104, 105]. The minimum projection removes orbital components below m and enforces occupation of site m , and both operations preserve a Slater determinant. The full-configuration measurement preserves the Slater form as well, because it collapses the state to occupied site orbitals. Both demons can therefore be propagated exactly on $M \times N$ matrices even though their conditioned histories differ.

1. The order-statistic projection

The Slater determinant $|\Phi\rangle = \phi_1 \wedge \dots \wedge \phi_N$ obeys

$$\Pi_{S,N}|\Phi\rangle = (P_S\phi_1) \wedge \dots \wedge (P_S\phi_N). \quad (\text{A2})$$

Its squared norm is $\det(\Phi^\dagger P_S \Phi)$. Taking S to exclude the first m sites gives $\det(\Phi^\dagger P_S \Phi) = \det(I - \Gamma_{A_m})$ by the determinant identity $\det(I - AB) = \det(I - BA)$.

Orthonormalize the surviving orbitals in Eq. (A2). Within their occupied subspace, choose an orbital χ_1 carrying the amplitude at site m and $N-1$ orbitals $\chi_2, \dots, \chi_N$ with zero amplitude there. For a nonzero probability outcome,

$$c_m|\Phi_S\rangle = \chi_1(m)\chi_2 \wedge \dots \wedge \chi_N, \quad (\text{A3})$$

up to the common determinant phase. The normalized state after occupation of m is therefore

$$|m\rangle \wedge \chi_2 \wedge \cdots \wedge \chi_N. \quad (\text{A4})$$

Zeroing rows below m , orthonormalizing, and forming the orthogonal complement of the site- m amplitude vector implements this map exactly. It preserves all coherences allowed by the minimum sector.

2. Rank-one sampling

For an orthonormal orbital matrix, the probability of a configuration is

$$p(\mathbf{n}) = |\det \Phi_{\mathbf{n}}|^2. \quad (\text{A5})$$

The sequential determinantal sampler selects a site with probability Γ_{jj}/N , projects the occupied subspace onto vectors vanishing at that site, and repeats in the remaining rank. Each ordering of a sampled configuration has probability $p(\mathbf{n})/N!$. The unordered sample thus obeys Eq. (A5) [103]. The recorded surprisal is $-\ln p(\mathbf{n})$, evaluated from the determinant of the original propagated orbitals at the selected sites.

3. Dephased order-statistic evolution

For two particles, complete position dephasing after minimum readout gives the wall-frame state

$$\rho_t = \sum_{d \geq 1} w_t(d) |0, d\rangle \langle 0, d|. \quad (\text{A6})$$

The transition probability from separation d to minimum m and final separation d' is

$$T_{m,d',d} = |U_{m0}U_{m+d',d} - U_{md}U_{m+d',0}|^2. \quad (\text{A7})$$

Conditional on the observed minimum alone,

$$p_t(m) = \sum_{d,d'} T_{m,d',d} w_t(d),$$

$$w_{t+1}(d') = \frac{\sum_d T_{m,d',d} w_t(d)}{p_t(m)}. \quad (\text{A8})$$

The record cost is $-\ln p_t(m)$ and the conditional potential-energy gain is

$$E_t = \Delta \left[2m + \sum_d d w_{t+1}(d) - \sum_d d w_t(d) \right]. \quad (\text{A9})$$

The conditional distribution retains the unresolved separation. Summing over m gives the same unconditional measurement-feedback map as a fully dephased position measurement whose configuration record is discarded. The two dephased measurements therefore give the same ensemble work at every cycle when they start from the same ensemble, while their record entropy rates remain different because they condition on different observations.

Appendix B: Exact bounds for the cycle

1. Measurement energy and stationary balance

Work first on M sites in the fixed- N sector with ordered basis $|n_1, \dots, n_N\rangle = c_{n_1}^\dagger \cdots c_{n_N}^\dagger |0\rangle$, $0 \leq n_1 < \cdots < n_N \leq M-1$. For nearest-neighbor hopping, the matrix element between configurations differing by one allowed hop is $-J$. The hop cannot exchange the ordering of particles. Decompose

$$K = \sum_{a=1}^N K_a, \quad (\text{B1})$$

where K_a moves the a th ordered coordinate by one site when the destination is empty. At fixed values of all other coordinates, the allowed values of n_a form an interval of L sites, with $L \leq M - N + 1$. The operator on this interval has eigenvalues $-2J \cos[k\pi/(L+1)]$, $k = 1, \dots, L$. Therefore

$$\|K_a\| = 2J \cos \frac{\pi}{M - N + 2} < 2J \quad (\text{B2})$$

for $N < M$, with zero norm when $N = M$. Equality follows by placing the other particles at the ends of the lattice, leaving the maximal interval for n_a .

Let A contain r ordered coordinates, and let P_q be the joint projectors onto their values. Every matrix element of K_a with $a \in A$ joins different measurement sectors. Every matrix element with $a \notin A$ remains within a sector. Hence

$$K - \sum_q P_q K P_q = \sum_{a \in A} K_a, \quad (\text{B3})$$

and its norm is at most $2rJ$. Any position-diagonal term V_d commutes with the projectors. For $H = K + V_d$ and arbitrary normalized σ ,

$$Q_{\text{meas}} = -\text{Tr} \left[\sigma \sum_{a \in A} K_a \right], \quad |Q_{\text{meas}}| \leq 2rJ. \quad (\text{B4})$$

The order-statistic measurement is $A = \{1\}$, for which Eq. (B2) is the exact finite-system norm. The semi-infinite bound follows directly from the norm bound for the open half-line blocks. The result depends on the projectors that define the measurement, not only on its outcome labels. Additional dephasing within a fixed minimum sector changes further hopping coherences and therefore contributes separately to the energy change.

For minimum feedback, let $\rho'_m = P_m \sigma P_m / p_m$. Since all sites below m are empty in this state, translating by m preserves the kinetic energy. It also preserves translation-invariant diagonal interactions, while

$$\text{Tr}(X T_m \rho'_m T_m^\dagger) = \text{Tr}(X \rho'_m) - Nm. \quad (\text{B5})$$

The unitary part preserves total energy. Averaging Eq. (B5) over outcomes therefore proves Eq. (11). The

wall occupies an empty site before and after relocation, so ideal changes of its on-site potential have zero mean work. The associated hopping coherences across the new boundary also vanish after the order-statistic projection.

In a stationary state with finite internal means, the endpoint differences average to zero. Combining $E = \Delta N \langle m \rangle$ with Eq. (15) gives $0 \leq \langle m \rangle \leq 2/(\alpha N)$. For a distribution p_m of mean μ , take $q_m = (1 + \mu)^{-1}[\mu/(1 + \mu)]^m$. Nonnegativity of $D(p||q)$ gives $H(p) \leq -\sum_m p_m \ln q_m = g(\mu)$, with the $\mu = 0$ limit understood continuously. The stationary chain rule and conditioning inequality yield

$$H_{\min} \leq H(m_t) \leq g(\langle m \rangle) \leq g[2/(\alpha N)]. \quad (\text{B6})$$

Finally, $\Pr(m \geq 1) \leq \langle m \rangle$ proves Eq. (19).

2. Bound on the minimum record

For a one-particle set S , let P_S be its one-body projector and let $\Pi_{S,k} = \wedge^k P_S$ project onto k -particle states supported entirely in S . Exterior multiplication gives the identity

$$\Pi_{S,N} c_f^\dagger = c_{P_S f}^\dagger \Pi_{S,N-1}. \quad (\text{B7})$$

The canonical anticommutation relation $\{c_g, c_g^\dagger\} = \|g\|^2 I$ implies $\|c_g^\dagger\| \leq \|g\|$ on every fixed-particle sector.

If $n_0|\psi\rangle = |\psi\rangle$, the state $|\chi\rangle = c_0|\psi\rangle$ has unit norm and $|\psi\rangle = c_0^\dagger|\chi\rangle$. A number-conserving quadratic unitary sends $c_0^\dagger$ to $c_f^\dagger$, $f = U|0\rangle$, and sends $|\chi\rangle$ to a normalized $|\chi'\rangle$. Equation (B7) then yields

$$\begin{aligned} \Pr(\text{all particles in } S) &= \|c_{P_S f}^\dagger \Pi_{S,N-1} |\chi'\rangle\|^2 \\ &\leq \|P_S f\|^2. \end{aligned} \quad (\text{B8})$$

No Gaussian assumption enters this step, so $|\chi\rangle$ may contain arbitrary correlations. A mixed state satisfying $n_0\rho = \rho$ has a decomposition into pure states in the occupied-origin subspace. Applying the inequality to each state and averaging preserves the same right-hand side. Taking $S = \{r, r+1, \dots\}$ gives Eq. (20).

For the one-particle Hamiltonian $h = k + \Delta x$, conservation of energy from $|0\rangle$ gives $\Delta\mu_1 = -\langle k \rangle_{U|0\rangle} \leq 2J$. The tail-sum identity $\langle m \rangle = \sum_{r \geq 1} \Pr(m \geq r)$ and the geometric maximum entropy bound then establish Eq. (21). Every exact order-statistic feedback outcome restores occupation of the origin, allowing the bound to be applied to the next history-conditioned state.

3. Decay of the offset kernel

If the post-feedback state is diagonal in position, the mean work of the next unitary step is a sum over the sites the particles start from,

$$E_{\text{diag}} = \sum_\ell e(\ell) \Gamma_{\ell\ell}, \quad e(\ell) \equiv \Delta \sum_j (j - \ell) |U_{j\ell}|^2, \quad (\text{B9})$$

where ℓ is the starting site measured from the wall and the offset kernel $e(\ell)$ is the mean energy a particle starting there gains in one cycle. We show that $e(\ell)$ decays factorially in ℓ at fixed cycle time.

Set $J = \hbar = 1$ in this appendix and let h_{bulk} be the tilted Hamiltonian on all integer sites. Removing the bond joining -1 and 0 gives $h_{\text{cut}} = h_{\text{bulk}} + v$, with $v = |-1\rangle\langle 0| + |0\rangle\langle -1|$. For an initial site $\ell \geq 0$, evolution under h_{cut} remains on the right half-line. Duhamel's identity gives

$$U_{\text{cut}}(\tau) - U_{\text{bulk}}(\tau) = -i \int_0^\tau U_{\text{cut}}(\tau - s) v U_{\text{bulk}}(s) ds. \quad (\text{B10})$$

The bulk propagator has modulus $|U_{j\ell}^{\text{bulk}}(s)| = |J_{j-\ell}[z(s)]|$. Thus

$$\begin{aligned} \|(U_{\text{cut}} - U_{\text{bulk}})|\ell\rangle\| &\leq B_\ell(\tau), \\ B_\ell(\tau) &= \int_0^\tau \sqrt{J_\ell[z(s)]^2 + J_{\ell+1}[z(s)]^2} ds. \end{aligned} \quad (\text{B11})$$

For a localized bulk initial state, the expectation of the shift operator, and hence of the bulk kinetic energy, is zero at all times. The half-line kernel obeys $e(\ell) = -\langle k_{\text{cut}} \rangle_{U_{\text{cut}}|\ell\rangle}$ by energy conservation. Since $\|k_{\text{cut}}\| \leq 2$, comparison of the two normalized evolved states yields

$$\frac{|e(\ell)|}{J} \leq 4B_\ell(\tau) + 2|J_\ell[z(\tau)]J_{\ell+1}[z(\tau)]|. \quad (\text{B12})$$

The second term is the expectation of the removed bond in the bulk state. The integral includes the entire history of the wave packet's exposure to the boundary.

For real z and integer $n \geq 0$, $|J_n(z)| \leq (|z|/2)^n/n!$. On $0 \leq s \leq \tau$, $|z(s)| \leq C = \min(2\tau, 4/\alpha)$, so

$$B_\ell(\tau) \leq \tau \sqrt{\frac{(C/2)^{2\ell}}{(\ell!)^2} + \frac{(C/2)^{2\ell+2}}{[(\ell+1)!]^2}}. \quad (\text{B13})$$

This proves factorial decay with offset at fixed cycle time. The locality estimate describes particles initially at specified sites, and by linearity the diagonal term in Eq. (B9). General coherent states also carry the off-diagonal work contribution in Eq. (28).

Appendix C: Construction of the single-particle reference

For one particle, feedback resets the state to $|0\rangle$, so

$$E_1(\tau) = \Delta \sum_j j |U_{j0}|^2, \quad H_1(\tau) = - \sum_j |U_{j0}|^2 \ln |U_{j0}|^2 \quad (\text{C1})$$

are deterministic. We construct the reference by sorting the sampled $\mathcal{P}_1 = JE_1/(\hbar\tau)$ values and taking the maximum $R_1 = E_1/H_1$ among all points at greater or equal power. For every sampled power we then retain

the largest R_1 among all points with equal or greater power, which implements Eq. (14) directly on the sampled curve.

Energy conservation gives $E_1 \leq 2J$. Consequently every feasible single-particle time at specified power $p > 0$ satisfies $\tau' \leq 2/\tilde{\mathcal{P}}$, where $\tilde{\mathcal{P}} = \mathcal{P}\hbar/J^2$. This provides a finite reference window for any positive-power comparison. The work-per-nat, barrier-error, and residual-phase comparisons use $M_1 = 300$, time step 0.01, and $\tau' \leq 200$, covering their feasible windows. The collective-gain map uses $M_1 = 160$, step 0.01, and windows of 800 for $\alpha \leq 0.2$ and 200 otherwise. These windows cover $2/\tilde{\mathcal{P}}$ for every colored cell in Fig. 3. A cell already below a sampled reference cannot become advantageous when additional single-particle times are admitted, since the reference envelope can only increase. We use this fact only for the very small powers whose complete feasible window extends beyond the displayed search range.

Long-time branches matter to low-power comparisons. At $\alpha = 0.3$ a branch near $\tau' = 105.7$ has $R_1 \simeq 0.794J$ per nat and $\mathcal{P}_1 \simeq 0.0139J^2/\hbar$, against a short-time maximum of $R_1 \simeq 0.747J$ per nat near $\tau' = 4.68$. The power constraint selects the appropriate branch. The reference at the main operating point is $R_{\text{ref}} = 0.59716J$ per nat. Continuous optimization near its maximum gives $\tau' = 8.94537$ and $R_1 = 0.59715572J$ per nat, a relative change below 2×10^{-7} from the 0.01 time grid. The corresponding single-particle power is $0.18255J^2/\hbar$, above the collective value.

Appendix D: Steady state and numerical convergence

1. Stationary trajectories and convergence

The initial state is the packed Fock configuration $0, \dots, N-1$. Each Gaussian cycle propagates the orbitals, computes the minimum law, samples the measured record, applies its projector, translates the frame, and orthonormalizes. Writing K_{cut} for the number of leading sites kept when the minimum law is evaluated, so that outcomes $m \geq K_{\text{cut}}$ are discarded, one Cholesky factorization of $I - \Gamma_{[0, K_{\text{cut}}]}$ supplies all leading determinants in Eq. (A1). A diagonal regularization of 10^{-13} permits evaluation at singular empty-interval matrices. Direct many-body comparisons quantify its probability error at about 10^{-13} .

The broad (α, N, τ) scan uses $M = 140$, $K_{\text{cut}} = 70$, 8000 cycles, one random seed, $0.1 \leq \alpha \leq 1.2$ in steps of 0.025, and $0.3 \leq \tau \leq 6$ in steps of 0.1. The displayed particle range is $1 \leq N \leq 16$. The low-entropy classification for $N > 10$, $\alpha > 0.4$ uses 4000 cycles per point and the resolved-set criterion $E > 0$, $H > 0.005$ nats per cycle. The first fifth of each trajectory is discarded. Every peak reported in Table III is then settled on its own, with four independent 15 000-cycle trajectories on $M = 200$, $K_{\text{cut}} = 110$ and cycle-time resolution 0.025.

The broad scan locates the maximizing particle number and the neighborhood of the cycle time, and the refined calculation fixes the value. Repeating every refined peak on $M = 280$, $K_{\text{cut}} = 150$ leaves all reported digits unchanged. Adjacent- N values are reported with their trajectory spreads, so the maximizing integer is interpreted with that resolution.

The work, coherence, dephasing, barrier-error, and residual-phase calculations use 352 trajectories totaling 4 296 000 cycles. Typical controls use two independent trajectories of 12 000 cycles with 2400 discarded. The main operating point and the recorded-misplacement runs use four trajectories of 15 000 cycles with 3000 discarded. The Gaussian controls use $M = 140$, $K_{\text{cut}} = 60$, and the main operating point uses $M = 200$, $K_{\text{cut}} = 110$. The maximum missing probability in the sampled minimum law is below 4×10^{-14} after regularized evaluation. Full-configuration states give $|E_{\text{coh}}| < 2 \times 10^{-15}J$.

The exact dephased two-particle calculation uses $M = 80$ up to $\tau = 4$ and $M = 120$ for longer times. Repetitions at $M = 120$ for $\tau = 1.85, 2.5, 4$ and at $M = 160$ for $\tau = 8$ change mean work and entropy by less than 2×10^{-12} in the chosen units. At $\tau = 8$, the maximum conditional probability in the largest ten separations at $M = 120$ is about 3.1×10^{-9} .

We compute finite-trajectory work with Eq. (8) and entropy from the actual conditional surprisal. Ratios are formed from pooled mean work and entropy. Quoted spreads are half the range of individual trajectory ratios, which measures the scatter between trajectories rather than the standard error of their mean. Block averages of the paired work and surprisal fluctuations provide an independent estimate of the sampling uncertainty. Stationarity in the sampled regime is supported by stable internal profiles, block averages, boundary occupation, and size comparisons. The theorems requiring stationarity explicitly assume finite stationary internal means. The stationary state is generated by the repeated many-body measurement and feedback cycle. The localized bulk Bloch solution is used only for the separate single-particle propagation problem.

2. Validation against finite many-body propagation

The many-body Hamiltonian is also constructed directly in the occupation basis with explicit fermionic signs. Direct propagation, projective measurement, and moving-frame work agree with the orbital representation. The correlated-state tail bound was tested on 60 mixed states in occupied-origin subspaces. The measurement energy identity and ordered-coordinate norm bounds were checked in 34 finite systems, including a nearest-neighbor density interaction. Maximum discrepancies in the operator-energy identities are below $3 \times 10^{-15}J$. The minimum probabilities differ by at most 1.1×10^{-13} , and post-projection state infidelity is below 10^{-15} . The dephased two-particle evolution agrees with direct many-

body density-matrix evolution to 2×10^{-16} in outcome probability.

Appendix E: Experimental parameters

1. Band and phase scales

For $V(z) = sE_r \sin^2(\pi z/d)$, a plane-wave basis with reciprocal indices G gives the dimensionless band matrix

$$H_{GG'}(q) = [(q+2G)^2 + s/2]\delta_{GG'} - \frac{s}{4}(\delta_{G,G'+1} + \delta_{G,G'-1}), \quad (\text{E1})$$

where $-1 \leq q < 1$. We use $-25 \leq G \leq 25$ and determine s from one quarter of the lowest-band width. At $\alpha = 0.15$ this yields $s = 10.30384$, $J_{\text{width}}/h = 524.2985$ Hz, and minimum gap 137.3518 kHz. The first Fourier hopping coefficient is $J_1/h = 524.1996$ Hz and the second is $J_2/h = -5.7284$ Hz. Thus the bandwidth calibration closely represents the nearest-neighbor model, with a quantified longer-range correction. Such longer-range hopping also changes the ordered-coordinate energy bound and can be included explicitly in an experimental Hamiltonian.

The 1064-nm lattice recoil calibrates the bands. The 671-nm probe recoil enters the scattering problem through its absorption and emission wave vectors. Equation (35) uses their axial difference and the frozen-well frequency. Multiplying it by the mean scattering count gives the mean added oscillator occupation in a harmonic approximation. The probability of interband excitation requires the actual motional state and trap spectrum.

2. Adaptive threshold search

For a conditional distribution $p_0, \dots, p_{K-1}$, an ordered threshold question partitions a contiguous interval $[i, j]$ at k into $[i, k]$ and $[k+1, j]$. Let $C(i, j)$ be the probability-weighted number of questions required for this interval. The optimal alphabetic search obeys

$$C(i, i) = 0, \\ C(i, j) = \sum_{m=i}^j p_m + \min_{i \leq k < j} \{C(i, k) + C(k+1, j)\}. \quad (\text{E2})$$

The expected number of questions is $C(0, K-1)$, with any nonzero tail treated as a final interval and refined as needed. The recurrence follows because every outcome in $[i, j]$ incurs the first question and then the optimal cost of one subinterval. It is the alphabetic coding problem [106], for which the optimal tree is also obtained in linearithmic time [107]. Its mean length is at least $H(p)/\ln 2$. The conditional distribution fixes both the sequence of thresholds and the cost of the record.

3. Finite-run statistics

For a packed initial state, $X_0 = N(N-1)/2$. Terminal imaging of $n_1^{\text{final}}, \dots, n_N^{\text{final}}$ in the final wall frame measures

$$W_L = \Delta \left[NS_L + \sum_a n_a^{\text{final}} - X_0 \right]. \quad (\text{E3})$$

Terminal sampling includes the residual quantum position variance. The internal-frame mean can also be computed conditionally to compare with this measurement. For the 512 independent 30-cycle runs quoted in the text, these estimates give 1.06757 J and 1.06836 J per cycle, respectively. The mean surprisal is 1.00207 nats per cycle, and their ratio using terminal imaging is 1.06536 J per nat. These are finite-run values from the prepared state. The stationary values are those of Eq. (24).

For independent preparations indexed by a , define $Z_a = W_{L,a}/\langle W_L \rangle - \mathcal{I}_{L,a}/\langle \mathcal{I}_L \rangle$, where $\mathcal{I}_{L,a}$ is the total surprisal recorded in that run. Propagating the joint fluctuations of work and surprisal to leading order gives the relative standard error

$$\frac{\delta R}{R} \simeq \sqrt{\frac{\text{Var}(Z_a)}{n_{\text{runs}}}}. \quad (\text{E4})$$

For a long stationary trajectory the corresponding variance also contains the correlations between cycles at nonzero separation. Block averaging estimates this quantity while retaining the correlation between work and surprisal. A finite-run comparison must also include the terminal record and the physical duration used by each engine.

Appendix F: Quantum statistics

For spinless fermions, an on-site density interaction is $n_j(n_j - 1) = 0$. A nontrivial interacting extension uses, for example, $V \sum_j n_j n_{j+1}$ or a longer-range density interaction. These terms commute with all position-sector projectors and preserve Eq. (15). Translation invariance also preserves the stationary balance. Gaussian closure and the quadratic-propagation tail bound then require a separate treatment of the interacting evolution.

Soft-core bosons and multicomponent fermions admit on-site Hubbard interactions. In the far-detuned strong-coupling regime, doublon motion arises through virtual tunneling with amplitudes of order J^2/U . In a tilted lattice the virtual denominators contain $U \pm \Delta$, and near a resonance the corresponding states participate directly. The same order-statistic feedback can then be combined with interaction-dependent transport [54, 66].

On an open one-dimensional nearest-neighbor chain, the Jordan-Wigner transformation relates hard-core bosons to spinless fermions while preserving occupation

projectors and the hopping Hamiltonian. Mapped initial states therefore give identical position-measurement histories and stored potential energies under the present

feedback. Off-diagonal bosonic one-body operators contain fermionic strings, and geometries with exchanges around loops require their own statistics analysis.

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